Forkwise — Project Report

A-ISPC 2026 · Hooman, Binesh School, Iran
Website: forkwise.online · Code: github.com/hoomanyyy/forkwise

Hello, this is the full report for Forkwise. Here I am going to explain everything about Forkwise.

What is Forkwise? What problem does it solve? Who needs this platform?

In 2018, scientists ran a study to find out how different results and predictions can be. To do this, they gathered 29 different groups. The scientists gave all of them the same question and the same problem to solve, but the prediction of each group was a little different, and the results were not equally significant. I mean that the results of 20 of those 29 teams were really different and meaningful, which means the results of the other 9 teams were not very meaningful. So this study proved that predictions are not always the same.

So this project is made for people who want to compare different answers, so that you can decide which prediction is stronger.

That is not all our project does! It can also find the best line: a linear regression function or a non-linear one. The dashboard of answers and numbers is a little complicated and difficult to understand because of its structure, but after a while it becomes simple and familiar.

How does it work?

There are at least three steps to get a result. First, the system reads the file and extracts its information and content. Then it summarizes the file and gives the summary to the model, so the model can make its decisions.

Second, Python cleans the data and runs a linear regression for each file and row. For each prediction, it calculates the p-value, the statistics and much other work that needs to be calculated. At the end, we have the result, and the backend returns it to the frontend in JSON format through an API, like:

```
{"Message": "ok"}
```

This is not the real result of our structure and API; it is just a simple example of what the API returns.

Coffee example

For this example, we have these columns: age, cups of coffee per day, cigarettes per day, exercise time, BMI and heart attack risk. Because I created the dataset myself, I already knew the result. Our question to our agents was: "Is coffee bad for the heart?" Our platform gave the question to 3 different models, 10 times (it can also be 100). Every model gave us a different result and prediction, but none of the 10 predictions found a meaningful effect, and the verdict was "no effect". The "no effect" verdict means that coffee does not have an effect. The main problem is smoking, which means that most people who smoke also drink more coffee.



Figure 1. The result page of the coffee test (Run #26): the question, the verdict and the summary numbers.

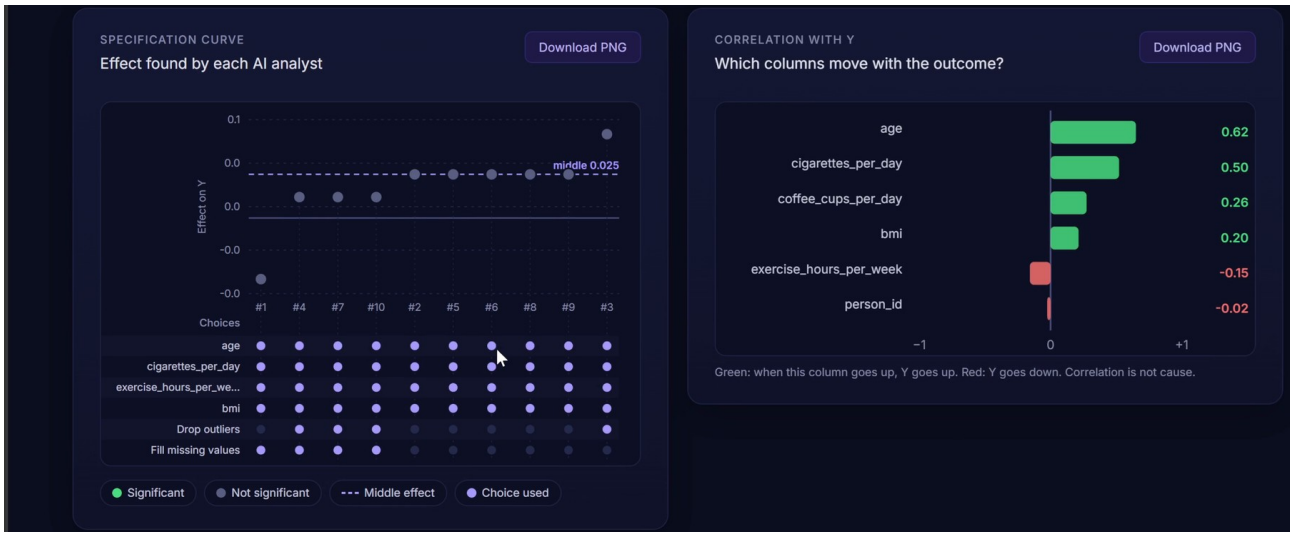


Figure 2. The specification curve (one dot for each AI analyst, with its choices below) and the correlation chart.



Figure 3. The data explorer: the real data with the best line and the formulas ranked by adjusted R-squared.

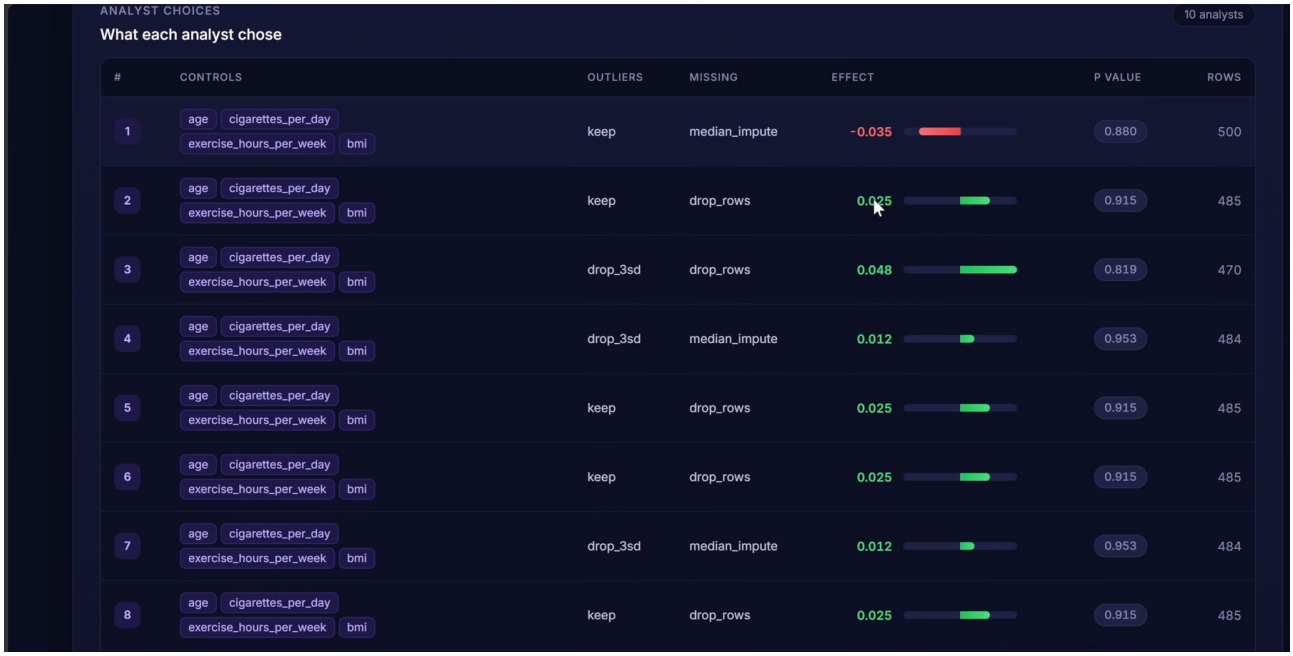


Figure 4. The choices of each AI analyst: controls, outliers, missing values, effect and p-value.

Tools I used

- I wrote the frontend with JavaScript and React.
- I wrote the backend with Python and FastAPI.
- I used Cloudflare Workers AI for the LLMs: 3 models that make the decisions.
- I used statsmodels, pandas and NumPy for machine learning and for working with numbers and files.
- I used openpyxl, xlrd and pandas to read file information.
- I used MySQL for the database because it is simple.

I also built documentation and help pages on the website, so people can learn more about the project and its structure.

Final words

But this project solves only a small part of people's problems, because knowledge is unlimited. When we learn more about one area of knowledge, we still cannot see its end. That is why we come into this world: to make work more comfortable for the next generation, and the next, and so on.

What I am trying to say is that I really want to solve one of people's problems, and I hope my project helps.